As you've already seen in previous chapters, contention between processors for access to a data structure can have a big performance impact. Something that scales nicely with the number of processors when that number is small may actually perform badly when the number of processors is much larger because of the huge increase in contention.

Consequently, when testing for the performance of multithreaded code, it's best to check the performance on systems with as many different configurations as possible, so you get a picture of the scalability graph. At the very least, you ought to test on a single-processor system *and* a system with as many processing cores as are available to you.

10.3. Summary

In this chapter we looked at various types of concurrency-related bugs that you might encounter, from deadlocks and livelocks to data races and other problematic race conditions. We followed that with techniques for locating bugs. These included issues to think about during code reviews, guidelines for writing testable code, and how to structure tests for concurrent code. Finally, we looked at some utility components that can help with testing.